

Proof. We argue by induction on n .

Base case $n = 1$. If $\lambda_k = 1$ for some index k and $\lambda_i = 0$ for $i \neq k$, then

$$\sum_{i=1}^n \lambda_i x_i = x_k, \quad \sum_{i=1}^n \lambda_i f(x_i) = f(x_k),$$

so Jensen's inequality

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i) \quad (\text{J})$$

holds with equality. This also covers the situation where some $\lambda_i = 0$.

Binary convexity. For two points $x, y \in I$ and a weight $t \in [0, 1]$ the defining property of convexity,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) \quad (\text{C})$$

is the elementary case from which the general inequality will be built.

Induction hypothesis. Assume that for a fixed integer $k \geq 2$ the statement (J) holds for any collection of k points and any non-negative weights summing to 1; i.e.

$$f\left(\sum_{i=1}^k \lambda_i x_i\right) \leq \sum_{i=1}^k \lambda_i f(x_i) \quad (\text{IH}_k)$$

whenever $\lambda_i \geq 0$ and $\sum_{i=1}^k \lambda_i = 1$.

Inductive step $k \rightarrow k+1$. Let $\lambda_1, \dots, \lambda_{k+1} \geq 0$ with $\sum_{i=1}^{k+1} \lambda_i = 1$. Set

$$\mu := \lambda_{k+1}, \quad t := 1 - \mu = \sum_{i=1}^k \lambda_i \ (\geq 0).$$

If $t = 0$ then $\mu = 1$ and we are in the base case, so (J) holds with equality. Hence we may assume $t > 0$.

Define the normalized convex combination of the first k points:

$$y := \frac{1}{t} \sum_{i=1}^k \lambda_i x_i.$$

Because the coefficients λ_i/t are non-negative and sum to 1, the induction hypothesis applied to the k points $x_1, \dots, x_k$ yields

$$f(y) \leq \frac{1}{t} \sum_{i=1}^k \lambda_i f(x_i). \quad (2)$$

Now apply the binary convexity inequality (C) to the pair (y, x_{k+1}) with weight t :

$$f(ty + \mu x_{k+1}) \leq tf(y) + \mu f(x_{k+1}). \quad (3)$$

Algebraic identity. Since $y = \frac{1}{t} \sum_{i=1}^k \lambda_i x_i$,

$$ty + \mu x_{k+1} = \sum_{i=1}^k \lambda_i x_i + \lambda_{k+1} x_{k+1} = \sum_{i=1}^{k+1} \lambda_i x_i. \quad (4)$$

Substituting (2) into the right-hand side of (3) gives

$$\begin{aligned} f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) &\leq t\left(\frac{1}{t} \sum_{i=1}^k \lambda_i f(x_i)\right) + \mu f(x_{k+1}) \\ &= \sum_{i=1}^k \lambda_i f(x_i) + \lambda_{k+1} f(x_{k+1}) \\ &= \sum_{i=1}^{k+1} \lambda_i f(x_i), \end{aligned}$$

which is precisely (J) for $k+1$ points. Thus the inductive step is proved.

Completion of induction. Since the inequality holds for the base case $n = 1$ and the inductive step is valid for every $k \geq 1$, by the principle of mathematical induction inequality (J) holds for all integers $n \geq 1$.

Edge cases. If some $\lambda_i = 0$, the corresponding term disappears from both sides; the remaining weights still sum to 1, so the argument above applies unchanged. If exactly one weight equals 1, the inequality reduces to equality, as shown in the base case.

Equality condition. Equality in (J) can occur only when every inequality used in the proof becomes an equality.

- In the binary convexity (C) equality holds iff either $t \in \{0, 1\}$ (i.e. one weight is zero) or f is affine on the line segment joining the two points. For a strictly convex f the latter forces the two points to coincide.
- In the induction hypothesis (IH_k) equality holds exactly when the same condition holds for the reduced set of points.

Putting these together, equality in Jensen's inequality holds *iff* one (or more) of the following is true:

1. **Weight concentration:** there exists j with $\lambda_j = 1$ and $\lambda_i = 0$ for $i \neq j$;
2. **All points equal:** $x_1 = x_2 = \dots = x_n$ (then the convex combination equals each x_i);
3. **Affineness on the convex hull:** the function f is affine on the convex hull of $\{x_1, \dots, x_n\}$. In that case the inequality reduces to an equality for any set of weights.

For a strictly convex f the only possibilities are (1) and (2). □